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CPSC 540-01: Statistical Machine Learning I

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Frequentist Vs. Bayesian Approaches

Hypothesis testing in statistical machine learning refers to the process of using inference methods that use data to support a particular hypothesis or theory. For example, one might collect data to support the theory that one party is faster than the other. Hypothesis testing differs to parameter estimation because rather than trying to predict a numerical value for the parameter, the process is instead providing evidence for or against a theory. This is often done by using null hypothesis significance testing, which relies on Reduction Ad Absurdum, a form of proof of contradiction. In other words, to prove that X is true, the process will assume that the opposite of X is true and provide evidence that it leads to something false, ridiculous, or highly improbable and, therefore, be evidence that X is true.

This homework assignment refers to some synthetic data that looks at different lifestyle and demographic variables and mental health/well-being for 500 people who are enrolled in Company X's wellness program. Some of the variables included are anxiety levels, average number of corgis pets per week, age, hair color, sex, household income, marriage status, etc. There are two questions we will analyze with the remainder of this paper. The first question deals with hypothesis testing, whereas the second deals with parameter estimation. Both questions will be addressed using both a Frequentist approach and a Bayesian approach.

Question 1: Hypothesis Testing - Difference of Joy Group Leadership

The first question we will address is as follows: Andrew and Betty are arguing about who is better at leading their Joy Group. Joy Groups were randomly assigned when employees joined the company. Is there evidence that Group A and Group B have the *same* anxiety score?

Frequentist Approach - Fisherian Hypothesis Testing

To answer this using a Frequentist approach, we will use Fisherian Hypothesis Testing, which asks how much evidence there is to support the null hypothesis. It does not require making a binary decision or having to either reject or fail to reject the null; rather, it is meant to observe if the data is consistent with the null hypothesis. We first decide on the null hypothesis: the mean anxiety scores for both Joy Groups are the same. Next, we will decide what test statistic we will use. For this question, we will use the group means for each leader's Joy Group and then calculate the difference between the two means. Finally, we must calculate the p-value and assess the significance of the p-value in reference to how much evidence we have with the null hypothesis.

In order to run this analysis, we did the following in the code. We fit a GLM model to see the effect of joy group membership on anxiety levels, so group membership is our predictor. Our null hypothesis, again, is that the effect is 0 or that joy group membership has no effect on anxiety levels. The GLM then produced a coefficient and a p-value (shown below) that we can assess in terms of significance.

```

## {r}
# fit the model according to the above
model <- glm(anxiety ~ group,
             data = data,
             family = gaussian())

# print out the summary stats
summary(model)

```

```

Call:
glm(formula = anxiety ~ group, family = gaussian(), data = data)

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.3830     0.2680  -1.429   0.154
groupB         0.1899     0.3868   0.491   0.624

(Dispersion parameter for gaussian family taken to be 18.66978)

    Null deviance: 9302.1  on 499  degrees of freedom
Residual deviance: 9297.6  on 498  degrees of freedom
AIC: 2886.4

Number of Fisher Scoring iterations: 2

```

In the figure above, we can see that the intercept is -0.3830. So, for Joy Group A, this represents the mean anxiety score. The coefficient for Joy Group B is 0.1899, which means that there is a difference of 0.1899 between Group A's mean and Group B's mean, with Group B's being slightly higher. The p-value for this difference is 0.624, which means the probability that we will observe another difference in means greater than or equal to 0.1899 is 0.624, which is more than half than likely. So, we could conclude that there is moderately low evidence against the null hypothesis or that there is no effect on the difference in mean anxiety scores. In other words, there is low evidence that the opposite would be true or of there being a difference in mean anxiety scores.

Bayesian Approach - Posterior Odds

To answer the same question, we will now use a Bayesian approach to hypothesis testing and look at posterior odds. When we look at the posterior odds, we compare the posterior probabilities of the two hypotheses (the null and the alternative) after we see the data to the prior probabilities before we see the data. Our first step is to define both the null and alternative hypotheses as follows:

- Null (H_0): Joy Group A and Joy Group B have the same mean anxiety scores.
- Alternative (H_1): Both groups have different anxiety scores.

The next step is to choose the prior distributions for the hypotheses before we see the data. Under the null hypothesis, we can represent the distributions as the same distribution because the null assumes there is no difference between the means. We will use a normal distribution with the mean at 0 and the variance as 5^2 , given that our data contains anxiety scores ranging from -10 and 10 and centered at 0. Under the alternative hypothesis, we will represent each group's mean as an independent, normal distribution. We will put this information into code by fitting two different GLM models. The first will include the group variable as a predictor in order to assess the effect of group membership on anxiety levels while the second will exclude the group variable as a predictor. Finally, we will compute the Bayes Factor and assess the impact of including the group variable as a predictor in our model.

```

{r}
#install.packages("brms")
library(brms)

# declare priors
priors <- prior(normal(0,5), class = b) +
  prior(cauchy(0,2), class = sigma)

# model 1: with group effect
# H1: group membership does have an effect on anxiety levels
fit1 <- brm(
  anxiety ~ group,
  data = data,
  family = gaussian(), # Normal distribution for continuous outcome
  prior = priors,
  save_pars = save_pars(all = TRUE),
  silent = 2, refresh = 0
)

```

```

{r}

priors_model2 <- prior(normal(0,5), class = Intercept) +
  prior(cauchy(0,2), class = sigma)

# model 2: without group effect
# H0: group membership does not have an effect on anxiety levels
fit2 <- brm(
  anxiety ~ 1, # Intercept-only model (no group effect)
  data = data,
  family = gaussian(),
  prior = priors_model2,
  save_pars = save_pars(all = TRUE),
  silent = 2, refresh = 0
)

```

```

{r}

# compute the bayes factor
bf <- bayes_factor(fit1, fit2)
print(bf)

```

```

Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
Iteration: 5
Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
Iteration: 5
Estimated Bayes factor in favor of fit1 over fit2: 0.10263

```

As shown in the figures above, the Bayes factor is calculated to be 0.10263, so there is a small preference for fit1 over fit2. This means that seeing the data observed makes the alternative hypothesis, H1, more plausible by a factor of 0.10263, which is not that much.

Question 2: Parameter Estimation - Direct Effect of Petting Corgis

The second question we will address is as follows: Assuming the DAG provided is correct and using covariates in a regression (GLM) model to adjust for variables, what is the direct causal effect of petting corgis on mental well-being?

Frequentist Approach - Confidence Intervals

To answer this question with a frequentist approach, we will use confidence intervals to see what the parameters are we can estimate using a GLM. So, we will fit a GLM with the mental well-being score as the outcome and the corgi pets, as well as other covariates according to the DAG, as predictors. According to the DAG, the covariates we should consider are age, hair color, and income. Then, using the output of the GLM, we will calculate the 95 percent confidence interval to assess uncertainty. For context, a frequentist confidence interval represents the expected proportion (long-term) of the interval that will contain a parameter, theta. In this case, the confidence interval will represent a range of plausible values that we think will represent the effect of corgi pets on mental well-being. If that confidence interval includes the value 0, then that means it is plausible for there to be an effect of 0 or no effect. If the interval does not include the value 0, then it is plausible for there to be an effect of corgi pets on mental well-being.

Frequentist Approach

As our frequentist approach, we will calculate the confidence interval for the output of a GLM.

```

{r}
# make sure our hair color variable is a factor
data$hair_color <- as.factor(data$hair_color)

# glm model
frequentist_model <- glm(mental_well ~ corgi + age + income_k + hair_color,
                          data = data,
                          family = gaussian())

# print glm output
summary(frequentist_model)

# confidence interval calculation
confint(frequentist_model, parm="corgi")

```

```

Call:
glm(formula = mental_well ~ corgi + age + income_k + hair_color,
    family = gaussian(), data = data)

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.04665	0.83423	-0.056	0.955427
corgi	0.39366	0.08776	4.486	9.05e-06 ***
age	0.05510	0.01293	4.262	2.42e-05 ***
income_k	0.04364	0.01254	3.480	0.000545 ***
hair_colorblonde	-0.52168	0.36347	-1.435	0.151841
hair_colorbrown	-0.28137	0.31651	-0.889	0.374459
hair_colorother	-1.04916	0.48943	-2.144	0.032552 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 8.504585)

Null deviance: 5321.8 on 499 degrees of freedom
 Residual deviance: 4192.8 on 493 degrees of freedom
 AIC: 2498.2

Number of Fisher Scoring iterations: 2

Waiting for profiling to be done...

2.5 % 97.5 %
 0.2216584 0.5656523

As shown in the figures above, the coefficient for corgi is 0.39366, which indicates that for each additional corgi pet, there is about a 0.394 increase in mental well-being, on average. The confidence calculated at the end of the output indicates the interval to be [0.2217, 0.5657]. This interval means that, with 95% confidence, we think the direct causal effect of petting corgis on mental well-being is within that range of values. This confidence level means that if we were to observe the long-term data, we would think that a 95% proportion would lie in this range. This range does not include the value 0, so we are pretty confident that there is an effect and that it is a positive effect. Therefore, we can conclude that petting corgis has a direct positive effect on mental well-being.

Bayesian Approach - Credible Intervals

To answer the same question using a Bayesian approach, we will use credible intervals. Credible intervals differ from frequentist confidence intervals in that they provide a range within which the parameter will lie with a certain level of posterior probability rather than looking at the proportion of long-term observances. We also now need to choose a prior distribution for the coefficient so that we can then estimate the posterior distribution. In our model, the outcome variable will still be mental well-being and the predictors will still be corgi pets, along with the three covariates we identified (age, income, and hair color).

For our prior distributions, we are going to use two different distributions. For the regression coefficients, we will use a normal distribution centered on a mean of 0 and with a variance of 1. This means that we are starting with the assumption that there is no effect, as our coefficients start near 0, and then we will allow for the data we observe to update those beliefs. For the priors of the standard deviation, we will use a Cauchy distribution centered at 0 and with a scale of 2. This means that there is moderate variability in what is expected for mental

well-being score values, but extreme variability is not expected. We are not using uniform priors to answer this question because they would actually make the assumption that we favor very large values. Instead, we are going to use weakly informative priors such as the normal and cauchy distributions that we just described.

```

##{r}
#install.packages("brms")
library(brms)

# declare priors
priors <- prior(normal(0,1), class = b) +
  prior(cauchy(0,2), class = sigma)

# glm model
bayesian_model <- brm(mental_well ~ corgi + age + income_k + hair_color,
  data = data,
  family = gaussian(),
  prior = priors,
  iter = 4000,
  warmup = 1000,
  chains = 4,
  cores = 4,
  seed = 123)

# output from glm model
summary(bayesian_model)

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI
Intercept	-0.13	0.83	-1.77
corgi	0.39	0.09	0.22
age	0.06	0.01	0.03
income_k	0.04	0.01	0.02
hair_colorblonde	-0.41	0.33	-1.06
hair_colorbrown	-0.19	0.29	-0.77
hair_colorother	-0.82	0.43	-1.66

	u-95% CI	Rhat	Bulk_ESS
Intercept	1.50	1.00	17941
corgi	0.56	1.00	16168
age	0.08	1.00	14271
income_k	0.07	1.00	16898
hair_colorblonde	0.25	1.00	14444
hair_colorbrown	0.37	1.00	15047
hair_colorother	0.01	1.00	16320

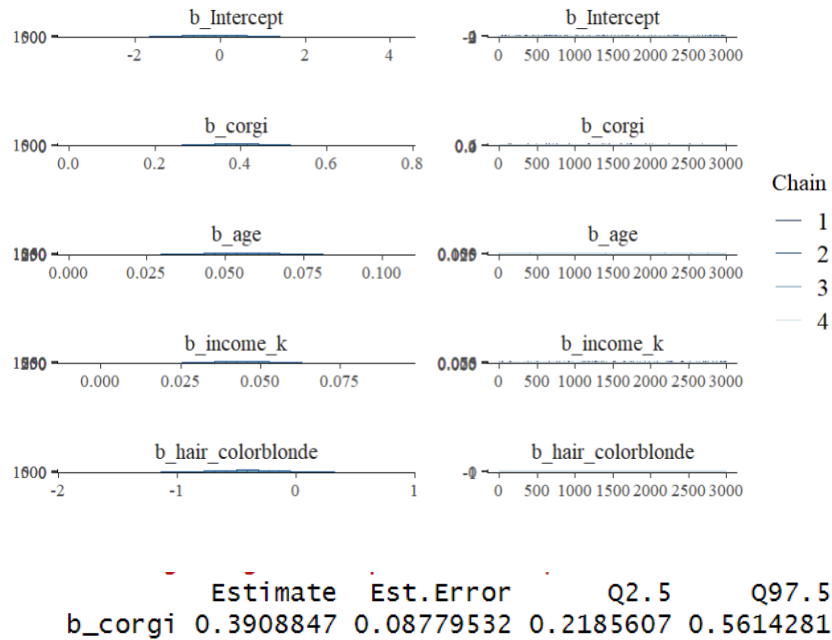
	Tail_ESS
Intercept	10522
corgi	9686
age	10141
income_k	9329
hair_colorblonde	9802
hair_colorbrown	10122
hair_colorother	9773

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI
sigma	2.92	0.09	2.74	3.11

	Rhat	Bulk_ESS	Tail_ESS
sigma	1.00	19050	9121

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



As shown in the figures above, the estimate for the posterior mean of corgi is 0.39, which represents that for every one-unit increase in the corgi variable (aka an additional corgi pet), there is a 0.39 increase in mental well-being. This is consistent with the estimate we got from our frequentist approach as well. In the last figure shows, we can also see the calculated lower and upper bounds of the credible interval at [0.22, 0.56]. This means that the 95% credible interval suggests there is a 95% probability that the actual effect of petting corgis on mental well-being is somewhere in the range of 0.22 to 0.56. Since this range does not include the value 0, which would indicate no effect, we can conclude that corgi pets have an effect on mental well-being.

So, we can conclude by using either a frequentist or Bayesian approach that petting corgis has a positive direct effect on mental well-being. We can also conclude that the effect is about a 0.39 increase in mental well-being, according to the scores we use in this study, for each additional corgi pet.

Discussion

Since we used both a frequentist and Bayesian approach to each of the two questions, we can quickly compare the results and conclusions of both approaches. For the first question, the frequentist approach resulted in observing low evidence against the null hypothesis or against there being a difference in mean anxiety scores. This does not conclude a binary decision, but it just shows how much evidence there is against the null. When we used the Bayesian approach of calculating the Bayes factor, we calculated it to be 0.10, which shows a small favor towards the alternative hypothesis. However, it is not by much, so we could still conclude that the null is pretty probable, and there is little evidence that the alternative hypothesis is true. So, for the first question, both approaches seem to come to a similar conclusion: there is weak evidence of there being an effect of Joy Group membership on mean anxiety levels. We could use the conclusions for the first question to prove that our random assignments of Joy Groups do not have an effect on anxiety levels. So, if someone were to complain and say that their Joy Group is either increasing or decreasing their anxiety level, on average, we can use the data analysis we computed to prove otherwise.

For the second question, the frequentist approach calculates the 95% confidence interval to be [0.22, 0.57]. So, the model predicts a 95% proportion of calculated effect values to be within that range if we were to observe long-term results. The Bayesian approach calculates the 95% credible interval to be [0.22, 0.56]. This means that we are 95% confident that the actual effect value is within that range. So, both of our approaches reach a very similar conclusion: there is a direct positive effect of petting corgis on mental well-being. We could use this conclusion to, perhaps, encourage corgi petting and corgi therapy as a method of improving mental well-being. So, if I were an exec at a university and I wanted to begin a program that

would help our student's mental health, I could recommend a corgi therapy program to my board of execs.

Personal Reflection

If I were to do this homework assignment again, I could make the following improvements. For one, I could consider including other frequentist or Bayesian approaches for both questions. So, I could compare, perhaps, two different frequentist approaches to two different Bayesian approaches and observe whether the approach used changes the conclusion we get. I could also do a more in-depth analysis. For example, for question 1, I could consider the effects of my covariates with my conclusion. So, does the benefit of petting corgis depend on age, income, or hair color? Overall, however, I think the analysis conducted in this homework assignment does an adequate job of showing the difference between frequentist and Bayesian approaches for both hypothesis testing and parameter estimation.

References

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